MATH 313 · Teaching Demonstration

Random Variables and Uncertainty

Probability Methods in Electrical and Computer Engineering

Dr. Ning Qi
Columbia University
Postdoctoral Research Scientist

A quick hello

Dr. Ning Qi

Columbia University
Postdoctoral Research Scientist
AI · Optimization · Energy

What I do

I work on AI and optimization for energy systems — applied math.

How I teach

I believe math should be easy, fun, and useful. You shouldn't feel dumb — you should feel curious.

Why you should care

Probability is the foundation of AI — the technology shaping our world. It also helps you in daily life: driving, shopping, teaching, voting.

Today's plan

25 minutes, 5 parts, 0 panic.

1 **Uncertainty is everywhere**

A weather forecast question to start.

2 **What is a random variable?**

The math behind uncertainty.

3 **How do we describe it?**

Distributions, ranges, quantiles.

4 **A real-world example: driving**

What does your 20% EV battery tells you.

5 **From uncertainty to decisions**

Three mindsets that help you act under uncertainty.

PART 1

Uncertainty is ***everywhere.***

Let's start with a question you've probably asked yourself.

A question to start…

What does this ***actually mean?***

"It will probably rain. "

"I bet it won't rain."

"70% of the day it will rain."

Think about 30 seconds.

Uncertainty is everywhere

Tomorrow's high

Forecast: 112°F.
Could feel like 118.
Could be only 105.

Your car's next month

Could run fine.
Could break down unexpectedly.

Commute to campus

El Centro → SDSU-IV.
Usually 20 min.
Sometimes 30 min.

Life is full of uncertainty. *Name for uncertainty— random variable*

PART 2

What is a

random variable?

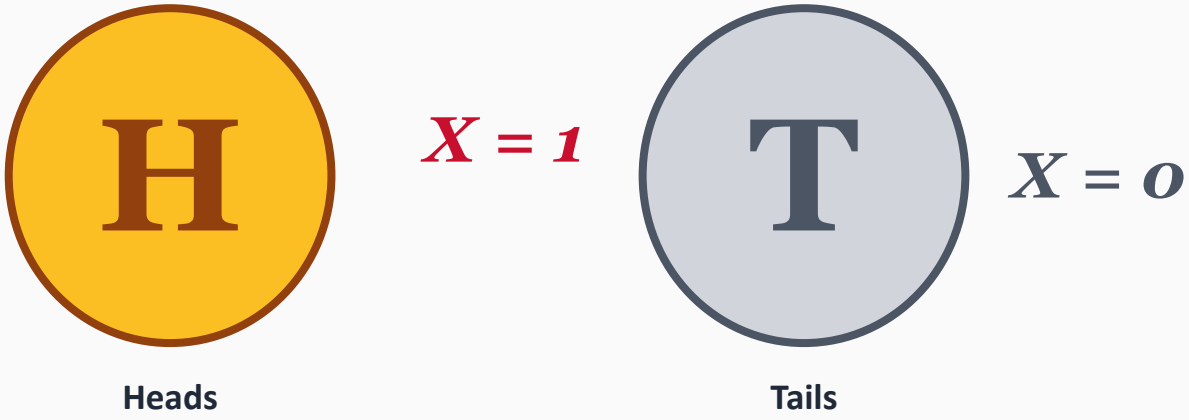
The math idea that makes uncertainty workable.

Everything you need to know about a random variable

Question: What are random variables?
Answer: Random variables are functions that translate random events into numbers.

Example: “Head” is a random event; “X” is number of heads

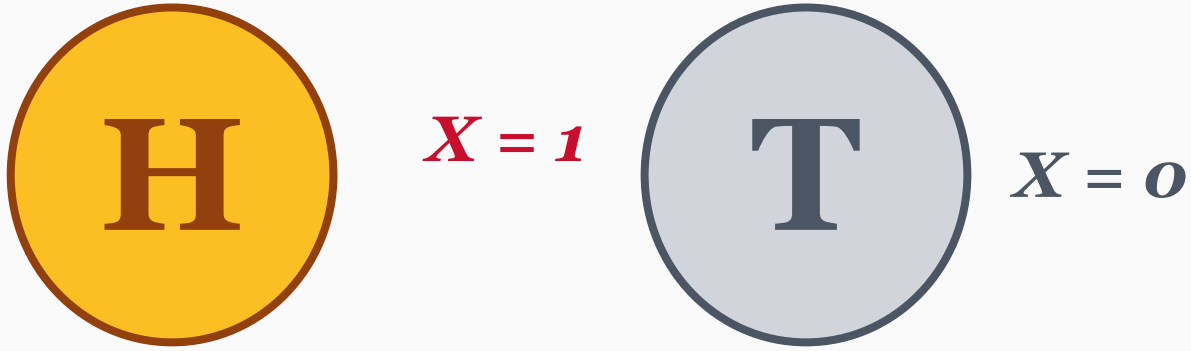
THE SIMPLEST EXAMPLE · Flip 1 coin



Everything you need to know about a random variable

Question: What is probability?
Answer: Likehood to each possible value of random variable.
Example: The Probability of “X =1” is 1/2

THE SIMPLEST EXAMPLE · Flip 1 coin



Heads

Tails

The probability of heads:

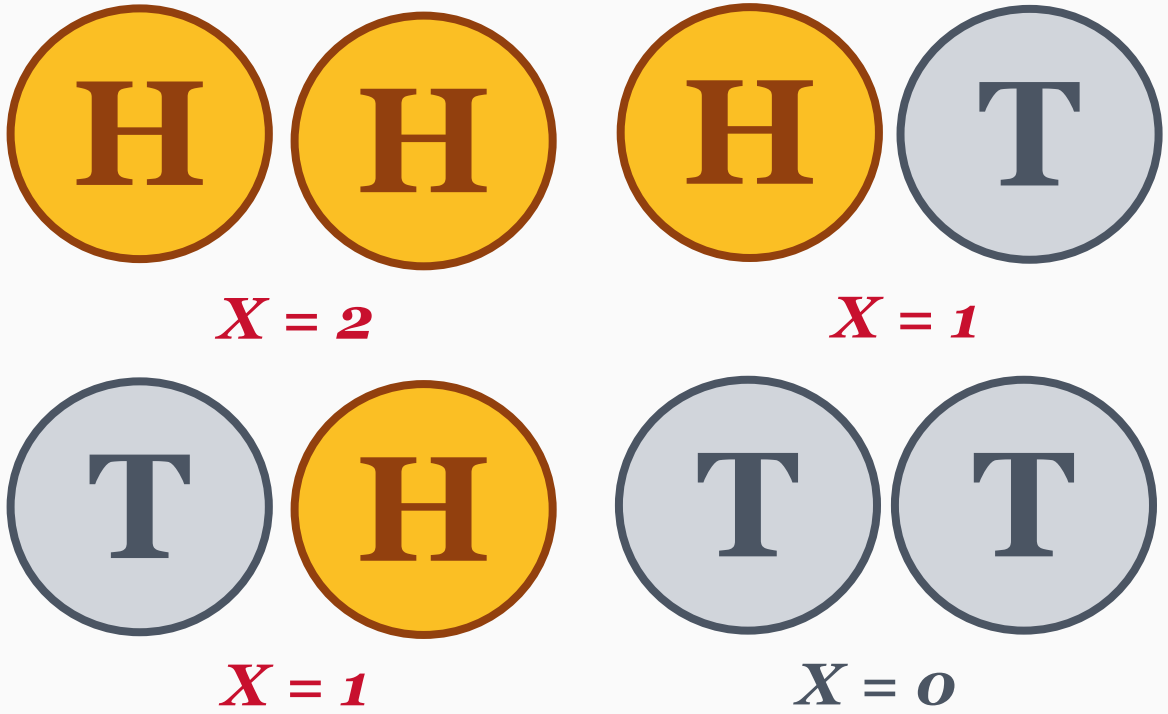
$$P(X = 1) = 1/2$$

1 favorable outcome (H) out of 2 possible.

Everything you need to know about a random variable

Question: Flip one coin is easy, how about two coins?

THE SIMPLEST EXAMPLE · Flip 2 coins



FLIP 2 COINS · X = number of heads

Look at each possible value of X:

X value	outcomes	P(X = x)
0	TT	1/4
1	HT, TH	1/2
2	HH	1/4

Three probabilities — one for each value of X.

PART 3

How do we

describe it?

Pictures first. Words second. Numbers — only when we need them.

From one probability to a distribution

Now flip 2 coins. Collect all the probabilities — and plot them.

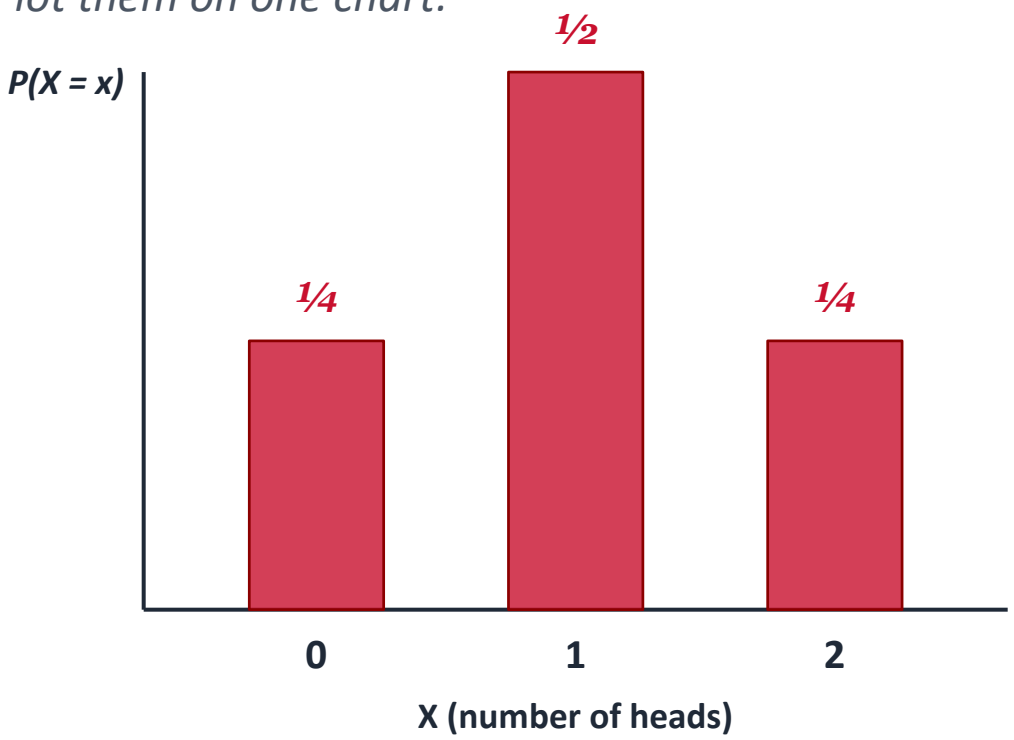
FLIP 2 COINS · X = number of heads

Look at each possible value of X:

X value	outcomes	P(X = x)
0	TT	1/4
1	HT, TH	1/2
2	HH	1/4

Three probabilities — one for each value of X.

Plot them on one chart:



This is a probability distribution.

Where do distributions come from?

Data (statistics)

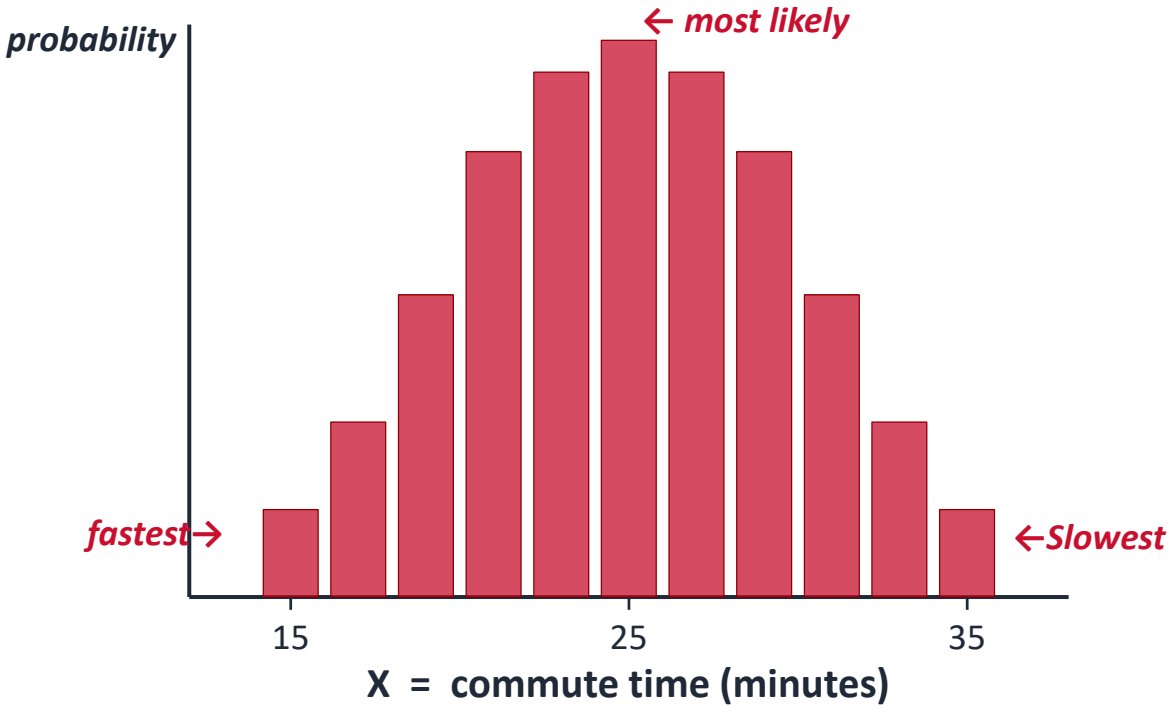
Collect many observations. Count them. Let the pattern emerge.

Way #1: a distribution

Imagine recording the same random variable many times.

Say X = commute time.

Record it every day for a year.



This shape is called

a distribution.

It tells us how likely each value is.

- Most days cluster around 25: **average**.
- A few days are really fast: **minimum**.
- A few are really slow: **maximum**.

The full picture isn't always available

Three common reasons — and why we need shortcuts.

Not enough data

Real-world data is limited.

365 drives to campus — nice.

But what if you only have 10? Or 3?

You can't draw an accurate picture from 3 points.

Privacy matters

We don't want to share everything.

Think about salaries at a company.

Not everyone wants their number posted online.

Share the big or vague picture — not every detail.

Too much to process

People need quick answers.

Showing 1,000 bars to a busy manager?

They want one number, fast — not a whole chart.

A summary helps people decide quickly.

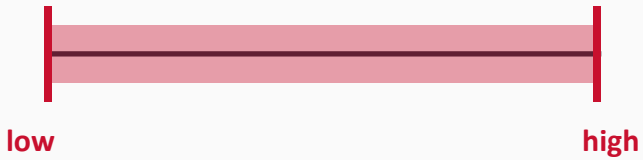
So we use shortcuts: ***a few key numbers that capture the big picture.***

Ways #2 & #3 & #4: bounds, quantiles, and average

Sometimes we don't need the whole distribution — just the big picture.

Range (Bounds)

"Between low and high."



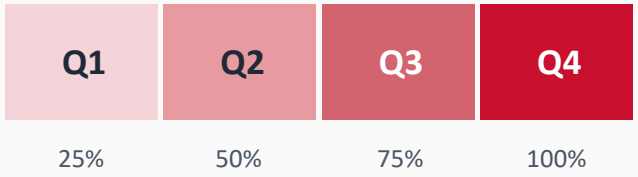
El Centro in July:

highs between 100°F and 118°F

→ a range of 18 degrees.

Quantiles

"Split into equal groups."



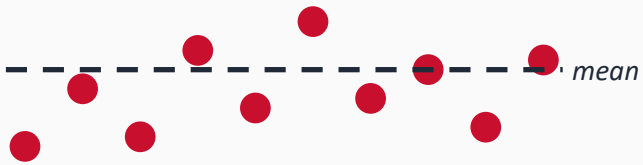
Quiz scores:

"Top 25% scored above 85."

→ the 75th quantile is 85.

Average (Mean)

"One number to summarize."



Drive to campus:

takes on average 25 minutes.

→ quick snapshot, hides variety.

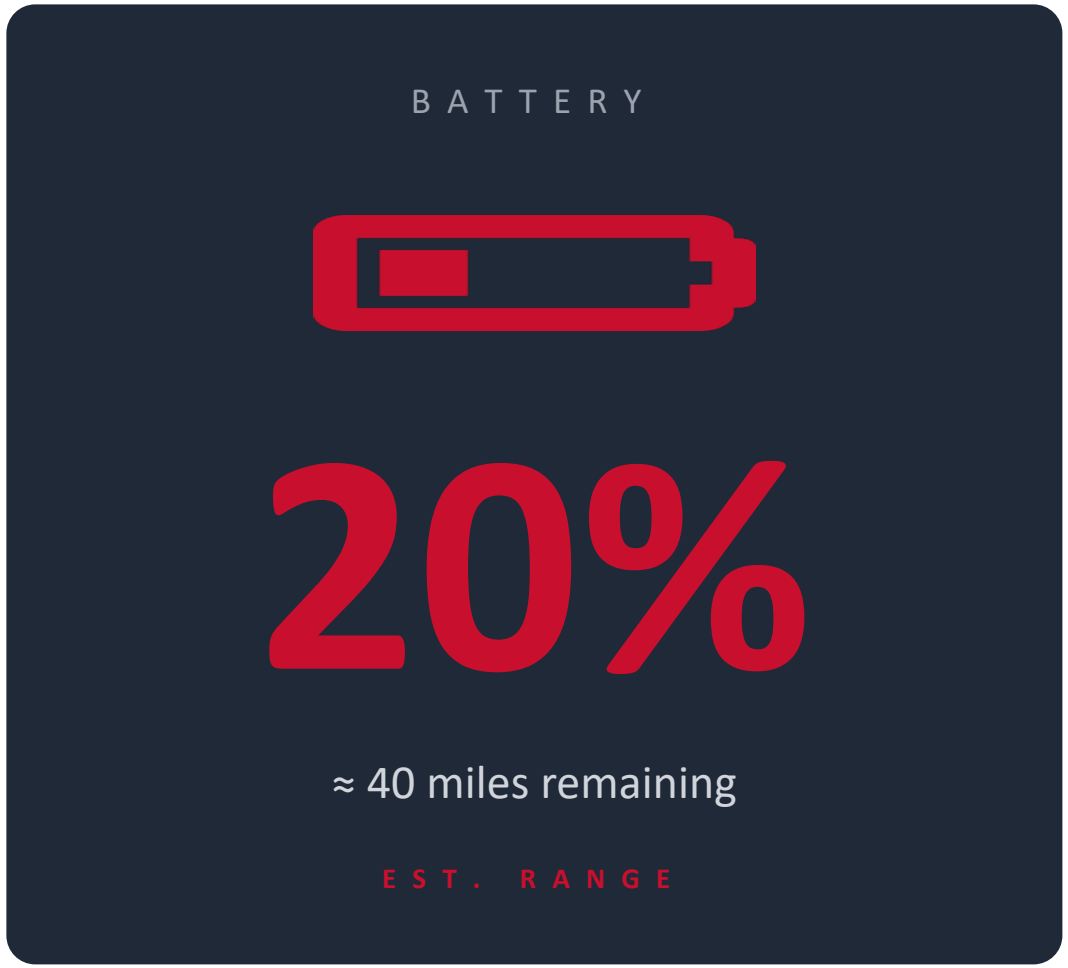
PART 4

A real-world example:

your electric car.

One distribution. Different conditions. Different places on the same curve.

Your car says: "20% battery left."



But here's the catch…

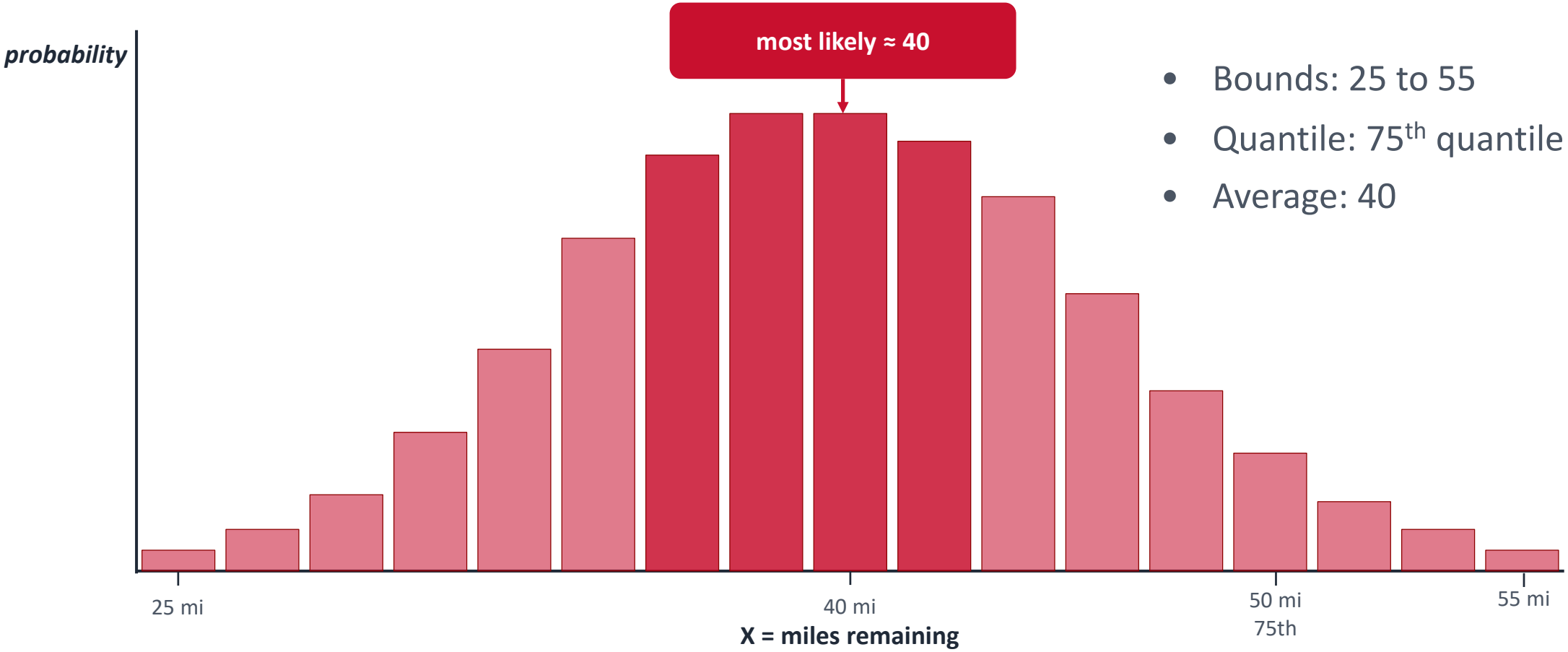
It's not really 40.

"40 miles" is an average.
The actual distance depends on:

- How fast you drive
- Outside temperature
- How old the battery is

What engineers actually see

Not one number — a whole range of possibilities:



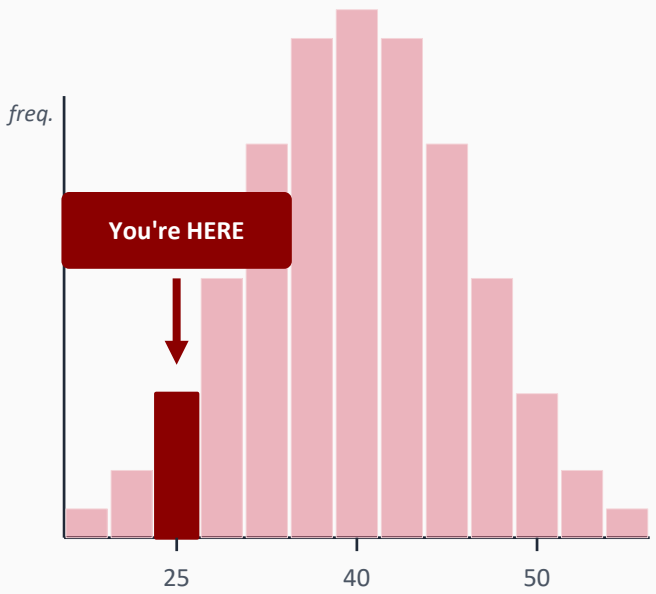
- Bounds: 25 to 55
- Quantile: 75th quantile is 50
- Average: 40

One distribution — where you land depends on conditions

Same EV, same 20%. The range is a random variable. Conditions decide where you are on the curve.

Hot + fast

110°F, driving 80 mph

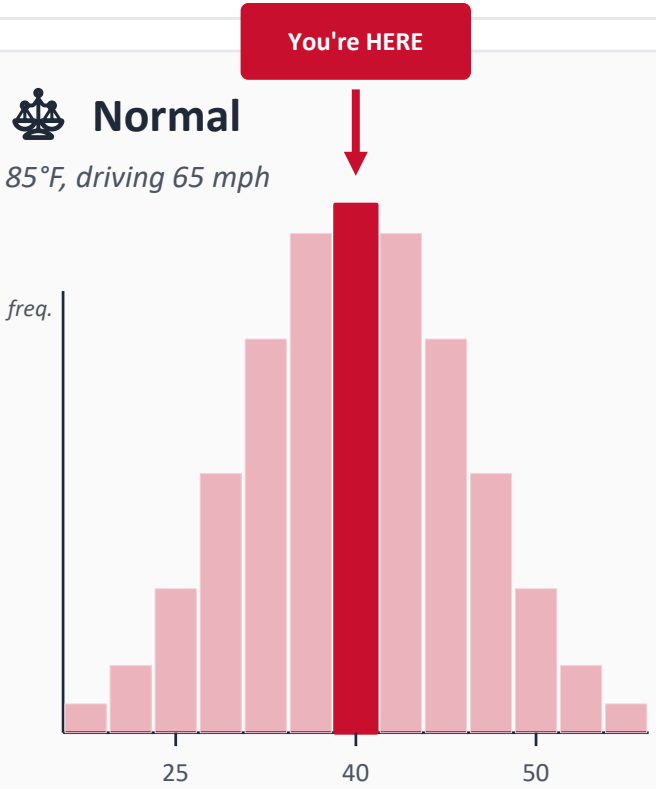


X = miles remaining

Range you'll likely get: ~25 mi

Normal

85°F, driving 65 mph

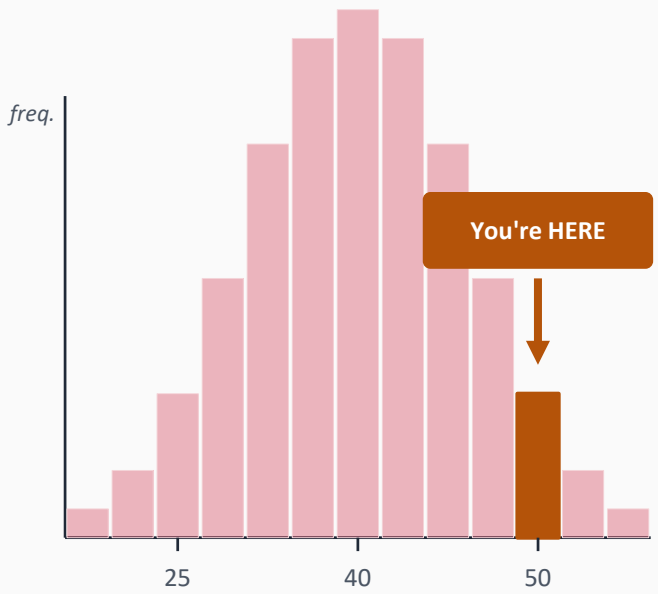


X = miles remaining

Range you'll likely get: ~40 mi

Cool + slow

65°F, driving 55 mph



X = miles remaining

Range you'll likely get: ~50 mi

It's all the same curve — conditions just tell you where you probably are on it.

PART 5

From uncertainty

to decisions.

Story #1: The lottery

Uncertainty often comes with risk.

Lottery

Millions of people buy tickets hoping to win.

Chance of winning the Mega Millions jackpot:

1 in 302,575,350

Expected profit per $2 ticket:

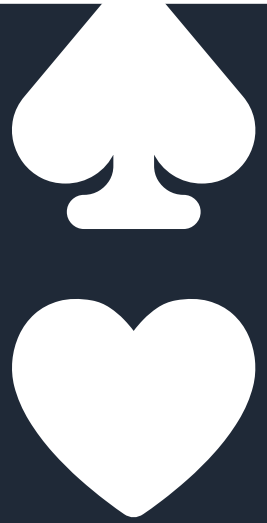
you lose about $1 on average.

RISK

Uncertainty that costs you.

Story #2: The MIT Blackjack Team

Uncertainty can also come with profit — if you know the math.



Blackjack

The MIT Blackjack Team · Movie 21 (2008)

MIT students used probability to track cards —

so they knew exactly when the odds were in their favor.

Expected profit per play:

positive — they won millions.

PROFIT

Uncertainty that pays you.

Three ways to decide under uncertainty

No style is "right" — each fits different situations.

Optimistic

Assume the best.

Plan for the good scenario. Fast, bold decisions. High reward if right — high cost if wrong.

"It'll probably be fine."

Robust

Prepare for the worst.

Plan for the bad scenario. Safer, more conservative. Lower risk — but you may "waste" effort.

"Just in case…"

Expected

Balance the odds.

Weigh each outcome by how likely it is. Pick the choice with the best average result.

"Do the math."

Back to your EV: 20% battery — what do you do?

The same 3 mindsets, applied to a decision you might actually face.

THE SITUATION

Your EV shows 20% (≈ 40 mi). Home is 40 mi away. Do you drive home, or stop to charge?

Optimistic

Trust the 40-mile estimate

Drive straight home at normal speed. Assume conditions are fine and you'll make it.

⚠ *If heat + speed drain battery faster → stranded on the 8.*

Robust

Plan for the worst case

Stop at the nearest charger, even 2 mi out of the way. Add 20 min — but guarantee safety.

⚠ *Costs extra time, even when it probably wasn't needed.*

Expected

Use the distribution

Slow to 55 mph → range distribution shifts right. Keep the nearest charger as backup.

✓ *Balances risk & time. Needs the car to tell you the range distribution.*

What we learned today

Uncertainty is everywhere

Weather, traffic, scores, batteries, solar — you can't escape it.

Random variable? Probability?

Random variable: translates random events to numbers. Probability: likehood to each possible value of random variable.

Distribution, range, quantile, average

Four tools to summarize X. Pick the right tool for the question.

Better decisions, not exact answers

Optimistic / Robust / Expected — each fits a different situation.

If you take this course — here's the full journey

1

Probability foundations

Set theory · axioms · conditional probability · Bayes' rule · independence

2

Random variables

PMF, PDF, CDF · mean, variance, moments · today's topic, in depth

3

Important distributions

Gaussian · Uniform · Exponential · Geometric — where they show up in real life

4

Multiple random variables

Joint & conditional distributions · correlation · covariance · jointly Gaussian

5

Estimation & decisions

Sums & long-term averages · signal detection · hypothesis testing · regression

6

Data science applications

Robust optimization · Bayesian decision theory · ML basics · random processes

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READ

Probability for Data Science — Stanley Chan (Purdue)

Free textbook + video lectures. Used at Purdue and beyond.

probability4datascience.com

READ

The Drunkard's Walk — Leonard Mlodinow

A fun non-technical book about how randomness shapes our lives.

Available at any library or bookstore

WATCH

The movie 21 (2008)

MIT Blackjack story — entertainment, not a how-to guide!

imdb.com/title/tt0478087

The best way to learn math is to use it.

Thank *you.*

Questions?

I'd love to hear what you think.

Dr. Ning Qi

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